

CUSTOMER & ORDER ANALYSIS

SQL Analysis Report | Day 01

1. Business Objective

The objective of this analysis is to examine customer and order data for an online food delivery platform and derive business-oriented insights related to customer activity, spending, restaurant performance, and city-wise order volume.

2. Data Overview

The analysis uses two related tables from the food_delivery database. Customer information is stored in the Customers table, while transaction details are stored in the Orders table.

Table	Key Information	Purpose
Customers	customer_id, name, city	Customer master data
Orders	order_id, customer_id, restaurant, amount, order_date	Order and transaction data

The Orders table is linked to Customers through customer_id using a Foreign Key relationship.

3. Analysis Performed

The following business analyses were performed using SQL:

Identify customers who have placed at least one order.

Calculate total spending for each customer and identify the top 3 customers by spending.

Identify orders placed within the last 7 days relative to the latest order date.

Identify customers who have never placed an order.

Identify the restaurant with the highest number of orders.

Identify Bengaluru customers whose total spending exceeds ₹1,000.

Calculate the total number of orders for each city.

Calculate the average order amount for each restaurant.

Identify customers who placed more than 5 orders.

4. SQL Approach

The analysis was performed in MySQL Workbench. The database and tables were created first, followed by data insertion and verification. SQL queries were then developed according to each business requirement.

Key SQL techniques used included:

JOIN and LEFT JOIN for combining related customer and order data.

SUM(), COUNT(), and AVG() for aggregation and measurement.

GROUP BY for customer, restaurant, and city-level analysis.

WHERE and HAVING for filtering records and aggregated results.

ORDER BY and LIMIT for ranking and selecting top results.

MAX(), subqueries, and INTERVAL for date-based analysis.

COALESCE() and IS NULL for handling customers without orders.

5. Key Findings

Based on the supplied dataset and query results:

Business Area	Finding
Top customer by spending	Swathi – ₹1,250.00
Highest-order restaurant	Meghana Foods – 3 orders
Bengaluru customers spending > ₹1,000	Divya and Swathi
City with highest order volume	Bengaluru – 5 orders
Average order amount – Meghana Foods	₹916.67
Customers with more than 5 orders	None in the supplied dataset

6. Data Validation & Issue Resolution

During Task 1, the query initially returned Error Code 1146 because the query referenced the table as Customer, while the actual table name was Customers.

The table name was corrected to Customers, after which the query executed successfully. This highlighted the importance of validating table names and database objects before executing SQL queries.

7. Business Insight

The analysis demonstrates how customer and order data can be used to identify high-value customers, understand restaurant order volume, compare city-level activity, and identify customers with no order activity. These insights can support basic customer engagement, restaurant performance, and operational analysis.

8. Conclusion

The Day 01 Customer & Order Analysis challenge was completed using MySQL. The analysis covered customer spending, order activity, restaurant performance, and city-level order trends.

This exercise strengthened practical SQL skills including relational joins, aggregation, filtering, ranking, date analysis, and NULL handling. The detailed SQL statements are maintained separately in the Queries file, while the Questions file contains the original business requirements.